

Antibodies coupled to lipophilic dyes enable high-density membrane labeling in super-resolution and expansion microscopy

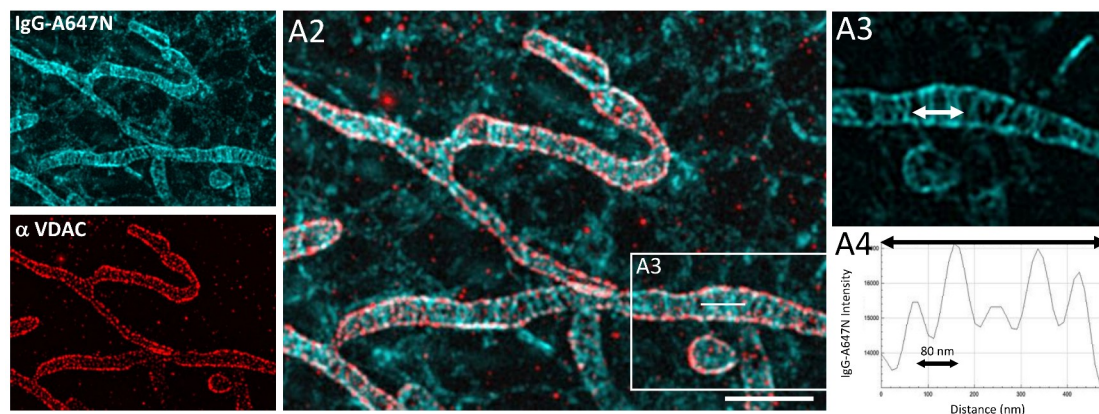
Jim DOMPIERRE, Silvia DEL POZO PERERA, Louis HURSON, Arnaud MOURIER,
Anne DEVIN, Manuel ROJO

¹ Institut de Biochimie et Génétique Cellulaires (IBGC), UMR 5095 CNRS – Université d Bordeaux
Bordeaux - FRANCE
(corresponding author: manuel.rojo@ibgc.cnrs.fr)

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ABSTRACT

Classical immunolabelling approaches can achieve homogeneous and continuous labeling of cellular membranes and organelles at wide-field and confocal resolution. In super-resolution and expansion microscopy, however, the lack of high-density labels hampers the localization of membrane proteins and protein complexes within their membrane context. We show that secondary antibodies coupled to lipophilic dyes ATTO 647N or ATTO 550 brightly label the nuclear envelope, mitochondria, and endoplasmic reticulum of fixed, permeabilized cells, and that graded labeling intensities allow selective visualization of organelles. State-of-the-art super-resolution and expansion microscopy reveal high-density labeling of nuclear and mitochondrial membranes, comparable to dedicated membrane-labeling approaches. We show that dye-conjugated IgG allow to resolve mitochondria-ER contacts and mitochondrial ultrastructure as well as precise visualization of the nuclear envelope and its invaginations. This study shows that secondary antibodies conjugated to lipophilic fluorophores represent stable, convenient and affordable tools for organelle visualization in conventional microscopy and for high-density labeling of membranes in super-resolution and expansion microscopy.



Expansion microscopy (ExM) reveals that IgG-ATTO 647N densely labels mitochondrial membranes. Fixed and permeabilized HeLa cells were decorated with goat-anti-rabbit IgG-ATTO 647N (IgG-A647N, cyan) and outer mitochondrial membranes were labeled with mouse anti-VDAC (red). Expanded gels were visualized by wide-field microscopy. A4 displays the intensity profile of cristae membranes along the line drawn in A3.

REFERENCE: <https://www.biorxiv.org/content/10.64898/2026.07.10.737767v1>